

Amirmohammad (Amir) Mohammadi (U.S. Permanent Resident)

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EDUCATION

Texas A&M University, College Station, Texas December 2026

Doctor of Philosophy in Computer Engineering

Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

Sharif University of Technology, Tehran, Iran

February 2021

Master of Science in Electrical Engineering

Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring

University of Tabriz, Tabriz, Iran

September 2018

Bachelor of Science in Electrical Engineering

Project: Development of a Data Logger for Motion Sensors with Python

PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas

September 2022 – December 2026

Graduate Research Assistant

- **Built** a few-shot whole slide image classification pipeline to investigate parameter-efficient fine-tuning of vision-language models for disease diagnosis in digital pathology applications.
- **Developed** a distribution-aware adaptation algorithm for Transformer-based foundation models as a parameter-efficient fine-tuning technique for waveform-based sensing applications.
- **Improved** the classification accuracy of a convolutional-based deep learning model by designing a time-frequency feature combination and engineering pipeline for audio-based applications.
- **Reduced** ground truth data requirements by a factor of 15 for physiological time-series signals by developing physics-informed neural networks for cuffless blood pressure measurement.
- **Disseminated** research findings through publications, presentations, and reports to funding agencies.

Sharif University of Technology, Tehran, Iran

July 2019 – February 2021

Graduate Student Researcher

- **Designed** an ECG + EDA wearable sensor with a 3× longer battery life compared to alternatives and delivered 94% mental stress detection accuracy across 18 participants using machine learning models.

SKILLS

- **Tools:** Python, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, MATLAB, C, Hugging Face, Transformers, Git, Linux, VS Code, Jupyter Notebook, High-performance Computing.
- **Interests:** Data Science, Applied AI, Foundation Models, Deep Learning Architectures & Techniques, Generative AI, Physics-Informed Machine Learning, Multi-Modal Learning, Efficient AI, AI Hardware.

PEER-REVIEWED PUBLICATIONS

- J. Ritu, **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peebles, "Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification," in *IEEE Access*, 2026. [[link](#)]

- **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peeples, "Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification," *IEEE International Geoscience and Remote Sensing Symposium*, 2026. [\[link\]](#)
- F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, and J. Peeples, "Neighborhood Feature Pooling for Remote Sensing Image Classification," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, March 2026, pp. 1393–1402. [\[link\]](#)
- **A. Mohammadi**, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine and J. Peeples, "Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks," *OCEANS 2025 Brest*, 2025, pp. 1-6. [\[link\]](#)
- **A. Mohammadi**, T. Kelhe, D. Carreiro, A. Van Dine, and J. Peeples, "Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models," in *OCEANS 2025 Brest*, 2025, pp. 1-6. [\[link\]](#)
- K. Sel, **A. Mohammadi**, R. I. Pettigrew, and R. Jafari, "Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation," *npj Digital Medicine*, vol. 6, no. 1, p. 110, June 2023. [\[link\]](#)
- **A. Mohammadi**, M. Fakharzadeh, and B. Baraeinejad, "An Integrated Human Stress Detection Sensor Using Supervised Algorithms," in *IEEE Sensors Journal*, vol. 22, no. 8, pp. 8216-8223, April 2022. [\[link\]](#)

MANUSCRIPTS UNDER REVIEW

- **A. Mohammadi**, A. Van Dine, and J. Peeples, "Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification". [\[link\]](#)
- **A. Mohammadi**, R. Khadka, K. A. Grant, M. L. Benton, and J. Peeples, "Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling".

LANGUAGES

English (proficient); **Farsi** (native/bilingual); **Azerbaijani Turkish** (native/bilingual)